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Application of IoT-Based Building Services in Smart Homes: Impact on Interior Design and User Experience

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ABSTRACT

The rapid advancement of Internet of Things (IoT) technology has transformed conventional residential environments into intelligent and responsive living spaces. Smart homes equipped with IoT-based building services enable automated control of lighting, HVAC systems, security systems, energy management, appliances, and environmental monitoring devices. These technological developments have created new opportunities and challenges for interior designers, as spatial planning, material selection, furniture arrangement, and user interaction are increasingly influenced by connected systems. This review paper investigates the application of IoT-based building services in smart homes and examines their impact on interior design practices and user experience. The study explores how IoT technologies enhance comfort, convenience, energy efficiency, accessibility, and personalization within residential environments. Furthermore, the paper analyzes the integration of smart lighting, intelligent HVAC systems, automated security solutions, smart furniture, and voice-controlled interfaces with interior spaces. The review highlights that IoT is transforming interior design from a static aesthetic discipline into a technology-driven approach focused on adaptive and user-centered environments. However, challenges related to privacy, cybersecurity, technological complexity, interoperability, and initial investment remain significant barriers. Future smart homes will require stronger collaboration between interior designers, architects, engineers, and technology developers to create sustainable, intuitive, and human-focused living environments.

Keywords: *Internet of things (IoT), smart homes, building services, interior design, user experience, smart lighting, home automation, intelligent environments*

INTRODUCTION

The concept of residential environments has undergone significant transformation due to advancements in digital technologies. Traditionally, homes were designed primarily around architectural form, spatial organization, functionality, and aesthetics. However, the emergence of Internet of Things (IoT) technology has introduced a new dimension where buildings can sense, communicate, analyze, and respond to occupant needs. The Internet of Things refers to a network of interconnected physical devices equipped with sensors, communication capabilities, and data processing systems that allow information exchange and automated decision-making. In residential environments, IoT enables various building services including lighting control, heating and cooling management, security monitoring, energy optimization, and appliance automation. Smart homes represent an evolution from conventional residential spaces into adaptive environments capable of responding dynamically to user behaviour and environmental conditions. Research on IoT-based smart homes highlights that these systems combine hardware infrastructure, communication networks, software platforms, and user interfaces to deliver automated residential services. [1,2]

The integration of IoT into interior design has introduced significant changes in the way residential spaces are planned and experienced. Interior designers are no longer concerned only with visual appearance, furniture arrangement, and material selection but must also consider sensor placement, connectivity requirements, automation interfaces, and technological adaptability. For example, smart lighting systems influence ceiling design, electrical planning, and spatial atmosphere; intelligent HVAC systems affect zoning and comfort strategies; and smart security systems influence entrance design and circulation planning. A recent systematic review on IoT integration in interior design identified that although IoT adoption in built environments is increasing, the relationship between IoT technology and interior design practice remains comparatively underexplored. The study emphasized the need for designers to understand technological systems to effectively integrate them into future spaces. Therefore, this review examines how IoT-based building services influence smart home interiors and how these technologies reshape user experience, sustainability, comfort, and future residential design practices.[3]

LITERATURE REVIEW

Evolution of Smart Homes and IoT Technologies

The development of smart homes has progressed from basic automated systems to complex interconnected ecosystems. Early home automation systems focused mainly on individual functions such as programmable lighting or security alarms. However, IoT technology has enabled integration between multiple systems, allowing communication between different devices and creating coordinated responses. Modern smart homes use environmental sensors, wireless communication networks, cloud computing, artificial intelligence, mobile applications and voice assistants. These technologies allow homes to monitor occupancy patterns, optimize energy consumption, and personalize environmental conditions. Sun and Li

(2021) identified that smart home research has expanded across infrastructure layers, communication technologies, data analytics, and user service applications. Their review emphasized that successful smart homes require integration between technological systems and human-centered services.[4,5]

Table 1: *Evolution of Residential Automation Technologies.*

Development Stage	Technology	Main Function
Traditional Homes	Manual controls	Basic operation
Home Automation	Programmable devices	Scheduled operation
Smart Homes	Connected devices	Remote monitoring
IoT Smart Homes	Interconnected systems	Automated decision-making
Intelligent Homes	AI + IoT	Predictive adaptive environments

IOT-BASED BUILDING SERVICES IN SMART HOMES

Smart Lighting Systems and Interior Design Impact

Lighting is one of the most significant interior design elements because it influences visual comfort, atmosphere, emotional response, and spatial perception. IoT-based smart lighting systems introduce new possibilities by allowing automated adjustment of brightness, colour temperature, and lighting scenes according to occupant requirements. Smart lighting systems commonly include occupancy sensors, daylight sensors, wireless controllers, smartphone interfaces and automated scheduling. From an interior design perspective, smart lighting changes the traditional approach toward illumination planning. Designers can create dynamic environments where lighting adapts according to activities such as working, relaxing, entertaining, or sleeping.[6,7]

Table 2: *Impact of IoT Building Services on Interior Design.*

IoT Service	Interior Design Impact	User Experience Benefit
Smart Lighting	Flexible ambience creation	Personalized environment
Smart HVAC	Improved spatial comfort	Thermal satisfaction
Smart Security	Modified entrance planning	Increased safety
Smart Furniture	Adaptive layouts	Convenience
Energy Monitoring	Sustainable design decisions	Reduced energy cost

IOT-BASED BUILDING SERVICES IN SMART HOMES

Smart HVAC Systems and Their Influence on Interior Design

Heating, Ventilation, and Air Conditioning (HVAC) systems are among the most important building services influencing indoor comfort and energy consumption. In conventional residential buildings, HVAC systems generally operate through manual adjustment or fixed schedules. However, IoT-based HVAC systems introduce intelligent monitoring and automated control by connecting sensors, smart thermostats, and building management platforms. IoT-enabled HVAC systems continuously collect environmental information such as indoor temperature, relative humidity, carbon dioxide concentration, occupancy levels, air quality parameters and user preferences. Based on this information, the system automatically adjusts heating, cooling, and ventilation conditions to maintain comfortable indoor environments while reducing unnecessary energy consumption.[8]

From an interior design perspective, smart

HVAC systems influence spatial planning, furniture arrangement, ceiling design, and material selection. Interior designers must consider sensor locations, ventilation pathways, equipment accessibility, and user interaction points during the design process. For example, occupancy-based HVAC control allows different rooms within a home to maintain independent environmental conditions. A bedroom may require different temperature settings compared with a living room or home office. Such zoning strategies support personalized comfort while improving energy efficiency. According to Javaid et al. (2022), IoT-based smart building systems improve operational efficiency by enabling real-time monitoring and automated decision-making. The integration of sensors and intelligent control mechanisms allows buildings to respond dynamically to environmental changes.[9,10]

Smart Security Systems and Interior Space Planning

Security is one of the most widely adopted applications of IoT technology in smart homes. IoT-based security systems integrate cameras, motion sensors, smart locks, access control devices, and alarm systems through connected networks. Unlike traditional security systems that operate independently, smart security solutions provide continuous monitoring and remote accessibility through smartphones or digital interfaces. Major components include smart door locks, video doorbells, motion detection sensors, indoor and outdoor cameras and automated alarm systems. The integration of smart security systems influences interior design by changing the way designers approach entrances, circulation spaces, and privacy zones. The challenge for interior designers is to integrate these technologies discreetly while maintaining aesthetic quality and user comfort.[11]

Smart Appliances and Connected Kitchen Design

The kitchen is one of the most rapidly transforming areas in smart homes due to the integration of connected appliances. IoT-enabled refrigerators, ovens, washing machines, and cooking devices allow users to remotely monitor and control household activities. These technologies influence kitchen planning by requiring additional electrical provisions, network connectivity, sensor integration and flexible cabinet design. Future kitchen interiors are expected to become adaptive environments where appliances communicate with users and respond automatically to household requirements.[12,13]

Smart Furniture and Adaptive Interior Elements

The integration of IoT technology has expanded beyond building services into furniture design. Smart furniture combines embedded sensors, connectivity, and automation to improve functionality and user experience. Smart furniture supports personalized living environments by adapting according to user requirements.[14]

Table 3: *IoT-Based Building Services and Interior Design Implications.*

IoT-Based Service	Technology Components	Interior Design Impact
Smart Lighting	Sensors, LEDs, controllers	Dynamic lighting atmosphere
Smart HVAC	Sensors, smart thermostats	Improved zoning and comfort
Smart Security	Cameras, locks, sensors	Modified entrance design
Smart Appliances	Connected devices	Flexible kitchen planning
Smart Furniture	Embedded sensors	Adaptive furniture layouts
Energy Management	Smart meters	Sustainable design decisions

IMPACT OF IOT-BASED BUILDING SERVICES ON USER EXPERIENCE

The primary objective of smart home technology is not only automation but improving the overall user experience. User experience in smart homes includes physical comfort, convenience, personalization, safety, accessibility, and emotional satisfaction. IoT technologies transform homes from passive environments into interactive spaces capable of understanding and responding to occupant needs.[15]

Enhanced Comfort and Personalization

One of the greatest advantages of IoT-based homes is the ability to personalize environmental conditions. Users can customize temperature preferences, lighting scenes, entertainment settings, privacy controls and security preferences. For example, a smart home system can automatically adjust lighting intensity, temperature, and music based on a user's daily routine. This personalization creates a stronger relationship between occupants and their living environment.[16]

Improved Energy Efficiency and Sustainability

Energy efficiency has become a major factor in smart home adoption. IoT systems allow continuous monitoring of energy consumption and identify opportunities for reducing wastage. Smart energy management systems can detect unnecessary appliance usage, optimize HVAC operation, monitor electricity consumption and encourage sustainable behaviour. According to Wilson et al. (2017), smart home technologies can contribute to energy conservation by providing users with greater awareness and control over their energy consumption.[17]

Accessibility and Inclusive Design

IoT technology provides significant benefits for elderly users and people with disabilities by improving accessibility. For elderly occupants, smart homes can provide increased independence and safety by reducing dependence on manual controls. Interior designers must therefore consider universal design principles while integrating IoT systems into residential spaces.

Table 4: *User Experience Benefits of IoT-Based Smart Homes.*

User Experience Factor	IoT Contribution
Comfort	Automated temperature and lighting control
Convenience	Remote operation of devices
Safety	Intelligent monitoring systems
Personalization	Customized environmental settings
Sustainability	Energy optimization
Accessibility	Voice and automated controls

CHALLENGES IN IOT-BASED SMART HOME INTEGRATION

Although IoT provides significant opportunities for smart interiors, several challenges affect widespread adoption.

Privacy and Data Security

Smart home devices continuously collect information about occupant behaviour, preferences, and routines. This creates concerns regarding personal data protection, unauthorized access and cybersecurity threats. Strong encryption and secure communication protocols are essential for protecting users.[18]

Interoperability Between Devices

A major challenge in smart homes is compatibility between devices manufactured by different companies.

Different communication standards may create problems such as limited device interaction, complex installation and difficult maintenance. Future smart homes require universal communication frameworks that allow seamless integration.

Cost and Technological Complexity

The initial cost of IoT-based systems remains higher compared with conventional building services. Challenges include installation cost, maintenance requirements, user learning difficulties and dependence on internet connectivity. Designers must balance technological advancement with affordability and usability.[19]

Table 5: *Challenges and Possible Solutions.*

Challenge	Impact	Possible Solution
Privacy concerns	User hesitation	Strong cybersecurity measures
Device compatibility	Integration problems	Common communication standards
High initial cost	Limited adoption	Affordable modular systems
Technical complexity	User difficulties	Simple interfaces

FUTURE TRENDS OF IOT-BASED SMART HOMES AND INTERIOR DESIGN

The future of residential environments is expected to move beyond automation toward intelligent, adaptive, and emotionally responsive living spaces. IoT technology will continue to transform smart homes by combining artificial intelligence, advanced sensors, digital modelling, and sustainable building strategies. The future smart home will not only respond to user commands but will also anticipate occupant requirements by learning behaviour patterns and environmental preferences.

For interior designers, this transformation represents a significant shift in design practice. Future interiors will require consideration of invisible technological infrastructures such as sensor networks, data communication pathways, automation systems, and intelligent interfaces while maintaining aesthetic quality and human comfort.[20]

Artificial Intelligence (AI) and Predictive Smart Home Systems

The integration of Artificial Intelligence with IoT is one of the most significant developments in future smart homes. While conventional IoT systems collect and transmit data, AI enables systems to interpret information, identify patterns, and make independent decisions. AI-based smart homes can analyze occupant behaviour patterns, daily schedules, environmental preferences, energy consumption habits and weather conditions. Based on this information, systems can automatically optimize indoor conditions. A smart home may learn that occupants usually return home at 6:30 PM and automatically adjust indoor temperature, lighting intensity, air quality and entertainment settings before their arrival. This transition from **responsive automation** to **predictive intelligence** will significantly influence future interior environments.[21]

Digital Twin Technology in Smart Interior Design

Digital Twin technology represents an advanced approach where a virtual model of a physical home is created and continuously updated using real-time sensor data. In smart homes, digital twins can represent spatial organization, HVAC performance, energy consumption, lighting behaviour and user interaction patterns.

Interior designers can use digital twins during both design and operational phases. During design development, digital twins allow designers to simulate lighting scenarios, furniture arrangements, thermal comfort conditions and Energy performance. After occupancy, the digital twin can monitor actual performance and recommend improvements.

Human-Centered Smart Interior Design

Future smart homes must focus not only on technological performance but also on human experience. Excessive automation without considering user behaviour may create environments that feel complicated or uncomfortable. Human-centered smart design focuses on simplicity, personal control, emotional comfort, accessibility and privacy. Interior designers will play an important role in ensuring that technology remains supportive rather than dominant. For example, a smart lighting system should provide automated control but also allow occupants to manually adjust settings when desired. This balance between automation and human control is essential for successful smart home adoption.

Sustainable Smart Homes and Energy Management

Environmental sustainability will remain a major driver of IoT-based building services. Smart homes can contribute to sustainability by monitoring and optimizing resource consumption. Future energy management systems may integrate smart meters, solar energy systems, battery storage, smart appliances and AI-based energy prediction. IoT technology enables buildings to become active participants in energy management rather than passive consumers. Smart homes can reduce unnecessary electricity consumption, optimize renewable energy usage, improve appliance efficiency and support low-carbon lifestyles.

Voice-Controlled and Natural Interaction Systems

Voice assistants and natural language interfaces are becoming increasingly common in smart homes. Future systems will provide more intuitive interaction methods where occupants can communicate with their environment naturally. These technologies will particularly benefit elderly users and individuals with mobility limitations by improving accessibility.[22]

Table 6: *Future Technologies in IoT-Based Smart Homes.*

Emerging Technology	Application in Smart Homes	Impact on Interior Design
Artificial Intelligence	Predictive automation	Adaptive environments
Digital Twins	Performance simulation	Data-driven design decisions
Voice Interfaces	Natural interaction	Reduced physical controls
Smart Materials	Responsive surfaces	Dynamic interiors
Renewable Energy IoT	Energy optimization	Sustainable living
Robotics	Automated assistance	Future lifestyle changes

RESEARCH GAPS AND FUTURE RESEARCH OPPORTUNITIES

Although IoT-based smart homes have received significant attention, several research gaps remain, particularly from an interior design perspective.

Limited Integration Between Technology and Interior Design Practice

Most existing studies focus on engineering aspects such as communication protocols, sensors, and automation performance. However, limited research examines how IoT technologies influence interior

aesthetics, spatial planning, material selection, furniture design and human perception. Future research should develop integrated frameworks combining technology and interior design methodologies.

Lack of User-Centered Evaluation Models

Many smart home studies evaluate technical performance but provide limited understanding of user experience. Future research should investigate emotional responses to smart environments, user acceptance, long-term behavioural changes and technology fatigue. Understanding human interaction will be essential for developing successful smart interiors.

Need for Sustainable IoT Design Approaches

While IoT systems improve energy management, they also create challenges related to electronic waste, device replacement and energy consumption of connected devices. Future studies should focus on sustainable IoT ecosystems using low-energy sensors, recyclable components and long-life technology solutions.

CONCLUSION

The application of IoT-based building services has fundamentally transformed the concept of residential environments. Smart homes are evolving from simple automated spaces into intelligent ecosystems capable of sensing, analysing, and responding to occupant needs. This review highlights that IoT technologies significantly influence interior design by changing the way spaces are planned, experienced, and managed. Smart lighting systems enhance spatial atmosphere, intelligent HVAC systems improve comfort, smart security solutions increase safety, and connected appliances create more efficient residential environments. The integration of IoT also improves user experience by providing personalization, convenience, accessibility, and energy awareness. However, challenges related to privacy, cybersecurity, interoperability, cost, and technological complexity continue to affect widespread adoption. Future smart homes will increasingly combine IoT, artificial intelligence, digital twins, and sustainable energy systems to create adaptive environments. Interior designers will play a critical role in this transformation by ensuring that technological advancement remains aligned with human needs, aesthetic values, and sustainable development principles. Therefore, the future of interior design lies not only in creating visually appealing spaces but also in designing intelligent environments that actively support occupant comfort, well-being, and sustainable living.

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